June 13, 2025

Dear editors,

We wish to submit an original research article entitled "Finding Hidden Functional States from EEG Based on Topological Features" for consideration by the Journal of Neural Engineering. The manuscript is 12 pages long and includes 9 figures and 1 table.

Subject area: Neuroscience methods.

In this paper, we elaborate on the recently presented state-detecting algorithm (SDA) – an automated technique for detecting functional stages in EEG recordings – and present an alternative approach to feature engineering based on a novel field of topological data analysis (TDA). We also propose qSDA – a unique feature selection framework based on the performance of SDA employed to pick a reasonable number of quality features from a comprehensive topological description of almost 20,000 characteristics. The study was inspired by the remarkable results of the previous work that used spectral features with a unique dataset of EEG recordings. Our work aims to assess the applications of TDA to the same data as it is known to yield interesting and interpretable results that reflect the spatial structure of the data.

The developments indicate that proposed topological methods effectively capture interesting patterns observed in the EEG and yield well-separated stage boundaries that reflect previous results but do not entirely duplicate them. The significance is supported by the contrastingly poor performance of the approach when applied to the surrogate data obtained by shuffling the epochs of the original EEG. This outcome suggests that TDA complements the spectral features and reveals new, previously unknown patterns enclosed in the spatial structure of the EEG, which may have interesting physiological interpretations leading to more robust functional stage detection. Moreover, substantial portions of the study focus on the proposed feature selection framework, which effectively removed the noisy features and appeared consistent with the information value analysis. As a result, we successfully deduced the brain regions that produced the most valuable data for the task.

This work primarily targets neuroscientists interested in data-driven EEG annotation, such as for sleep or epilepsy studies, as well as researchers examining the brain mechanisms of meditative practices. Although prior studies have developed a robust mechanism for this task, our research focuses on fundamentally different aspects of the source data. Thus, the study presents the readers with new insights into the spatial structure of the EEG and describes a powerful mechanism applicable to similar tasks from other fields. We believe our findings have wide practical applications and essential theoretical interpretations that advance our understanding of human brain activity.

The results discussed in the manuscript have never been presented to the scientific community. We confirm that it has not been published elsewhere and is not under concurrent consideration by another journal. My co-authors and I do not have any conflict of interest that might influence the research. All authors have approved the manuscript and agree with its submission to the Journal of Neural Engineering.

Sincerely yours,

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